

CSE370 – Database Systems
Assignment 03
Summer 2023

1. Consider the following relation:

Project_Grade_Upgraded(Project_ID, Student_ID, Student_Name, Semester, Course, Section, Project_Title, Grade, Submission, Bonus)

Additional dependencies are as below:

FD1 - Project_ID → Project_Title

FD2 - Student_ID → Student_Name, Semester

FD3 - Submission → Bonus

FD4 - Course, Section → Grade

Give reasons to explain if this is in 1NF or not. If not, decompose it to 1NF.

Give reasons to explain if this is in 2NF or not. If not, decompose it to 2NF.

Give reasons to explain if this is in 3NF or not. If not, decompose it to 3NF.

2. Consider the following relation:

Books (AuthorName, BookTitle, Publisher, Price, Year, AuthorAffiliation, Type, Category)

The primary key is underlined.

The schema has the following additional Functional Dependencies:

AuthorName → AuthorAffiliation

BookTitle → Publisher, Price, Type

Type → Category

Identify and explain which normal form the above schema is in. Step by step normalize it until 3rd normal form. You MUST show the complete “current” schema after each normalization step.